

Master

CSE370 Databases Systems: Class Notes

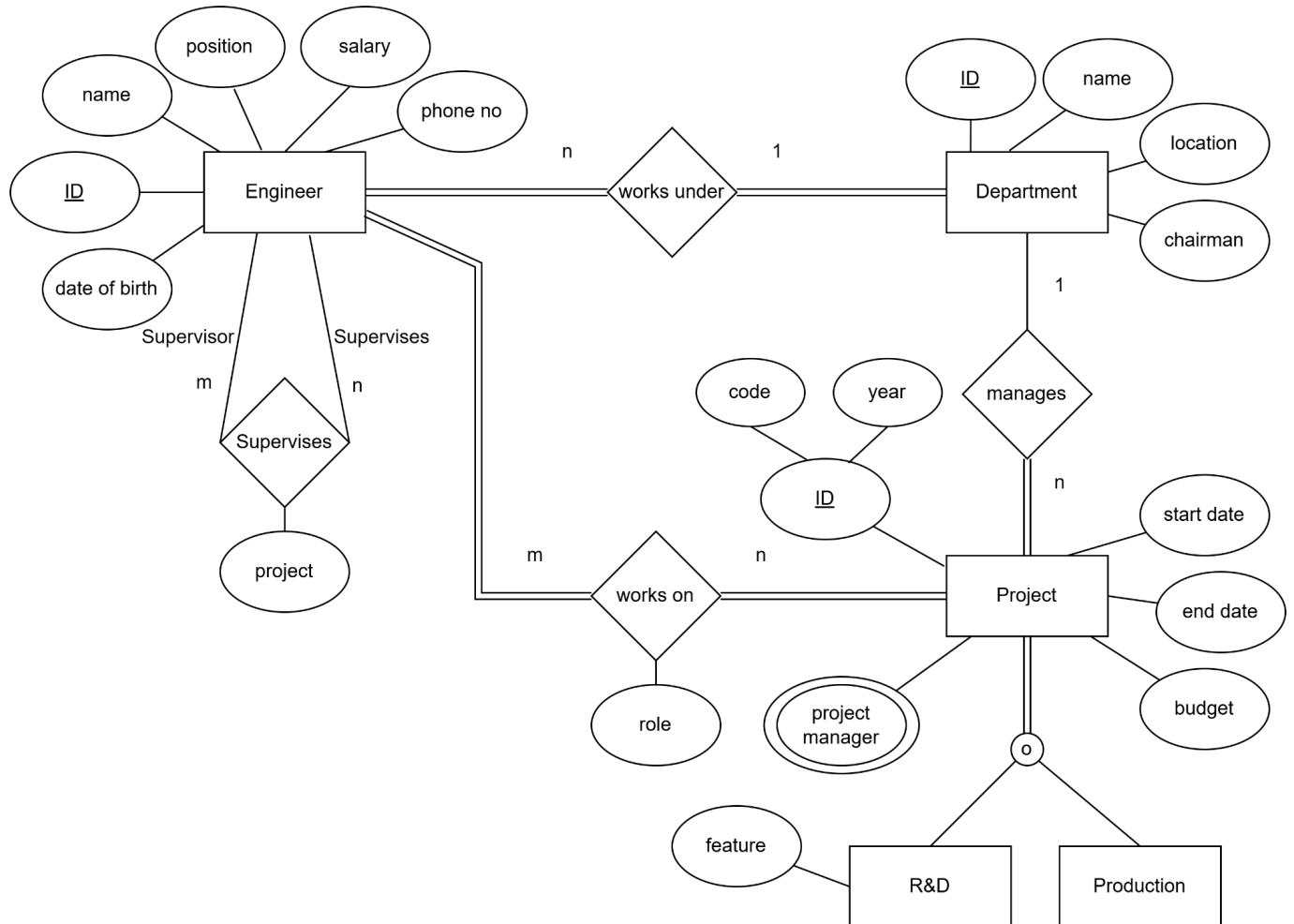
Spring 2026: Section: 07, 08, 09, 13

Class No.	Topic	Handout	Slide	Practice Sheet
1	Introduction to Databases		 Lectur...	 Practi...
2	Introduction to Databases (Wrap-up) Entity - What is it? Recording:  Lecture-2-Intro-Recap[MW].mp4	Whiteboard-1	ER	PS-2
3	ER Diagram Recording:  Lecture-3-ER-Diagram[ST].mp4  Lecture-3-ER-Diagram[MW].mp4	Whiteboard-2		
4	EER Diagram  Lecture-4-EER-Diagram[ST].mp4	Whiteboard-3	EER	PS-3
5	Database Constraints - I	Whiteboard-4		
6	Database Constraints - II	Whiteboard-5	Database Constraints	PS-4
7	Schema Diagram - I	Whiteboard - 6.1		
8	Schema Diagram - II	Whiteboard - 6.2	Schema	PS-5
9	[NOT IN MIDTERM] Normalization  Lecture-9-Normalization.mp4	Whiteboard-7	Normalization	PS-6
10	Review  [S26]Midterm_Review.mp4	Whiteboard		

Practice Schema:

Construct a relational Schema by mapping the following EER diagram (Figure Q3). The following conditions must be fulfilled when constructing the schema:

- For mapping the 1:N or 1:1 relationships, you **MUST** use the foreign key approach.
- For mapping the superclasses/subclasses, **select** the applicable option between 8C or 8D and **explain** briefly why the unselected option is not applicable for the given EER.



Class Schedule 13/07

Class No.	Date	Topic [Section 13, 07]
1	Feb 1, 2026	Lecture 1: Introduction to Databases
2	Feb 3, 2026	Lecture 2: Introduction to Databases (Wrap-up) Entity - What is it?
3	Feb 8, 2026	Lecture 3: ER Diagram
4	Feb 15, 2026	Lecture 4: EER Diagram
5	Feb 17, 2026	Lecture 5: Database Constraints
6	Feb 22, 2026	Lecture 6: Database Constraints Quiz 1: Lecture 1 - 3
7	Feb 24, 2026	Lecture 7: Schema Diagram - I
8	Mar 1, 2026	Lecture 8: Schema Diagram - II
9	Mar 3, 2026	Lecture 9: Normalization
10	Mar 8, 2026	Lecture 10:
11	Mar 10, 2026	Quiz 2: Lecture 4 - 8
12	Mar 15, 2026	Lecture 12: No Class

Class Schedule 08/09

Class No.	Date	Topic [Section 08, 09]
1	Feb 2, 2026	Lecture 1: Introduction to Databases
2	Feb 9, 2026	Lecture 2: Introduction to Databases (Wrap-up) Entity - What is it?
3	Feb 14, 2026	Lecture 3: ER Diagram
4	Feb 16, 2026	Lecture 4: EER Diagram
5	Feb 18, 2026	Lecture 5: Database Constraints
6	Feb 23, 2026	Lecture 6: Database Constraints Quiz 1: Lecture 1 - 3
7	Feb 25, 2026	Lecture 7: Schema Diagram - I
8	Mar 2, 2026	Lecture 8: Schema Diagram - II
9	Mar 4, 2026	Lecture 9: Normalization
10	Mar 9, 2026	Quiz 2: Lecture 4 - 8
11	Mar 11, 2026	Lecture 10:
12	Mar 16, 2026	Lecture 12: No Class

Assignment 1

Assignment 1

Q1. [2+2 Marks] Select if the following tasks will require DDL(Data Definition Language) or DML(Data Manipulation Language).

Explain your answer briefly (at max 2 sentences) with mention to schema and state:

- Rename the column called “StudentID” into “SID”
- Change the “class time” for section 7 to “01:00 PM - 2:05 PM”

- Ans: DDL, changes schema, not state
 - Ans: DML, changes state, not schema
-

Q2. [3+3 Marks] There are two types of data independence in database management systems: Logical Data Independence, and Physical Data Independence. Given the scenarios below, Identify what type of data independence both are and Explain your answer briefly (at max 3 sentences) with reference to Three-Schema Architecture.

- You work for a social media company. The company has decided to move from cloud hosting to physical hardware to support the server due to increase in demand.
- You work for a social media company. The company has decided to track the number of hours a user has stayed on their platform. This tracked_hours is not visible to the user but kept in the database.

Ans: First Case, PDI, changes in internal schema that does not affect conceptual schema.

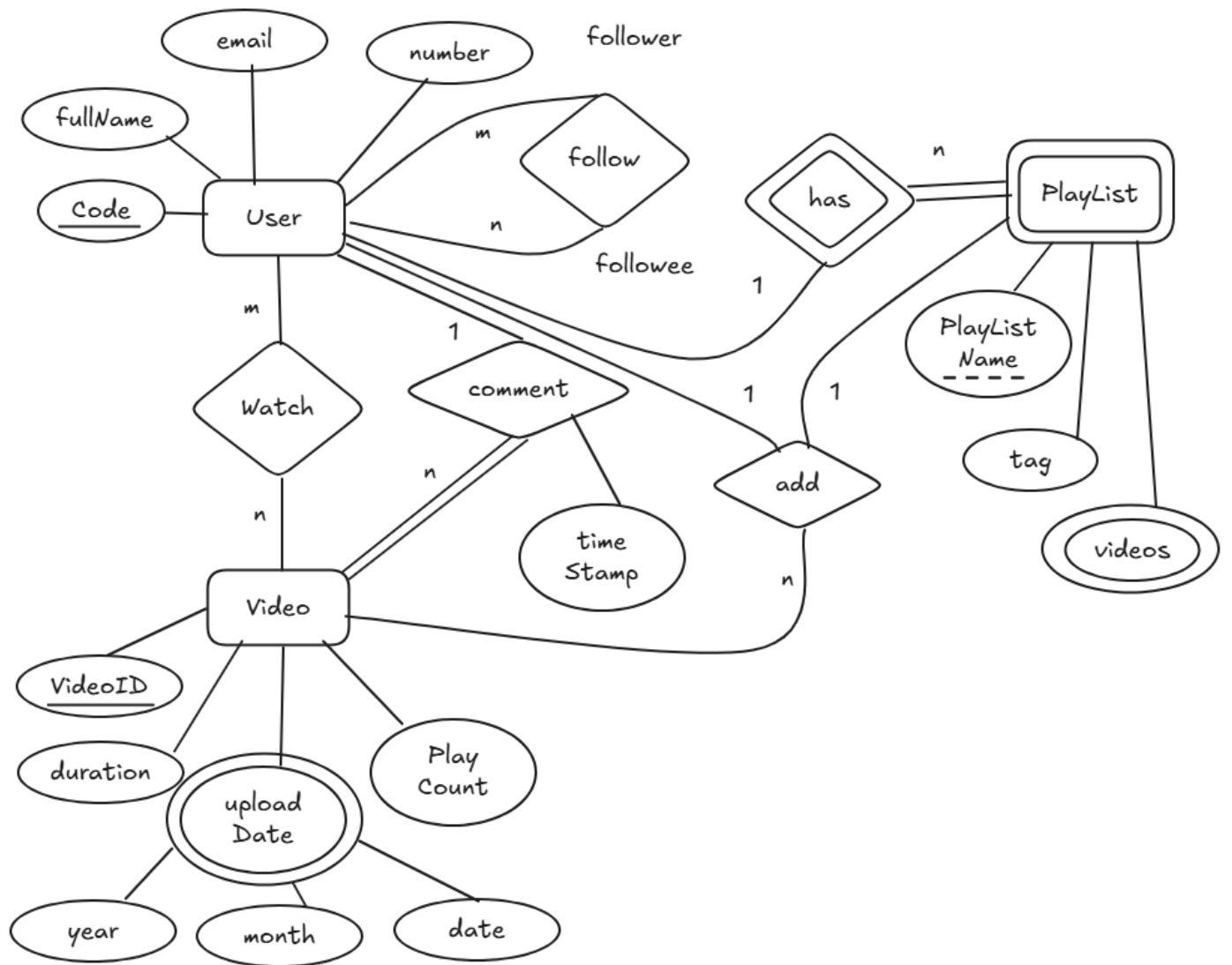
Second Case, LDI, changes in conceptual schema without affecting the External view.

Q3. [10 Marks] You are tasked with designing a Video Streaming System. The system must meet the following requirements:

- Each user will have a unique code, full name, email, and mobile number.
- Users can watch videos, which have a unique video ID, duration, upload_date, and play count.
- A video can be watched by many users and a user can watch many videos.
- Users can also comment on videos. Whenever a comment is made, the timestamp is recorded. A single user can comment multiple times.
- All videos must have comments, but not all users will leave a comment on the videos.
- Users can also follow other users.
- Some users might create a playlist. Each playlist has a playlist_name, tag, and a list of videos. Playlist_name is not unique, rather it is kept track of with the user's code and playlist_name combined.
- Users add videos to their playlist. One user can add many videos to their single playlist.

Now, draw an **ER diagram** of this system. Show at least 1 composite-multivalued attribute.

Write down all assumptions (if any).



Quiz 1

Quiz 1

Q1. There are two types of data independence in database management systems: Logical Data Independence, and Physical Data Independence. Read the scenario below, then answer the following 3 questions.

You work for a music streaming company. The company has decided to add a new column in its database called Location where it will track each user's country. This information is not shown in the front end.

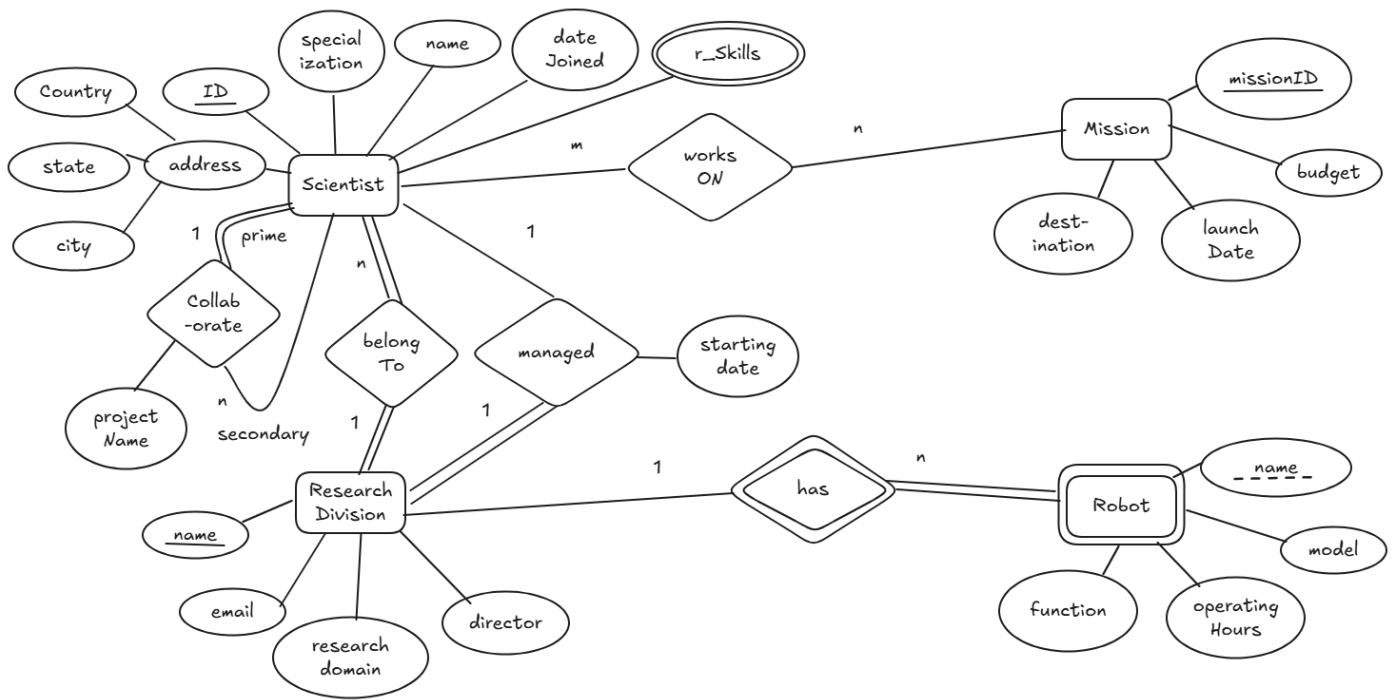
- What type of Data Independence is this? **(1 Mark)**
- Which Schema does this change in relation to Three Schema Architecture? **(1 Mark)**
- Name the next step of the affected schema you mentioned in (b) and mention whether it will show any changes due to the given scenario. **(1 Mark)**

- Ans: LDI
- Ans: Conceptual Schema
- Ans: External View, no changes

Q2. A Space Research Organization wants to create a database system for managing its missions, scientists, and facilities. The requirements are given below. **(7 Marks)**

- Scientists are part of a Research Division. Scientists have unique id, specialization, name, address, date_joined, and a list of research_skills.
- A Scientist's address is composed of country, state, and city.
- Each Research Division has a unique name, email, research_domain, and director.
- Each Scientist must belong to a Research Division, and each Research Division must have many scientists.
- Each Research Division is managed by one Scientist. A Scientist cannot manage more than one division. When a Scientist starts managing a Division, the starting_date is recorded.
- Scientists can collaborate with other Scientists. Each Scientist can collaborate with more than one colleague. A project name is recorded for each collaboration.
- Scientists also work on Space Missions. Each Mission has a unique mission_id, destination, budget, and launch_date.
- Some Research Divisions have Robots for assisting research. Each Robot has a name, model, function, and operating_hours. None of these attributes are guaranteed to be unique, different divisions may even have robots with the same name. However, note that the same Division will not reuse the same name for its own robots.

Now, draw an ER diagram of this system. Write down all assumptions (if any).



Assignment 2

Assignment 2

Q1. [2+2 Marks] Refer to the table below to answer the following questions:

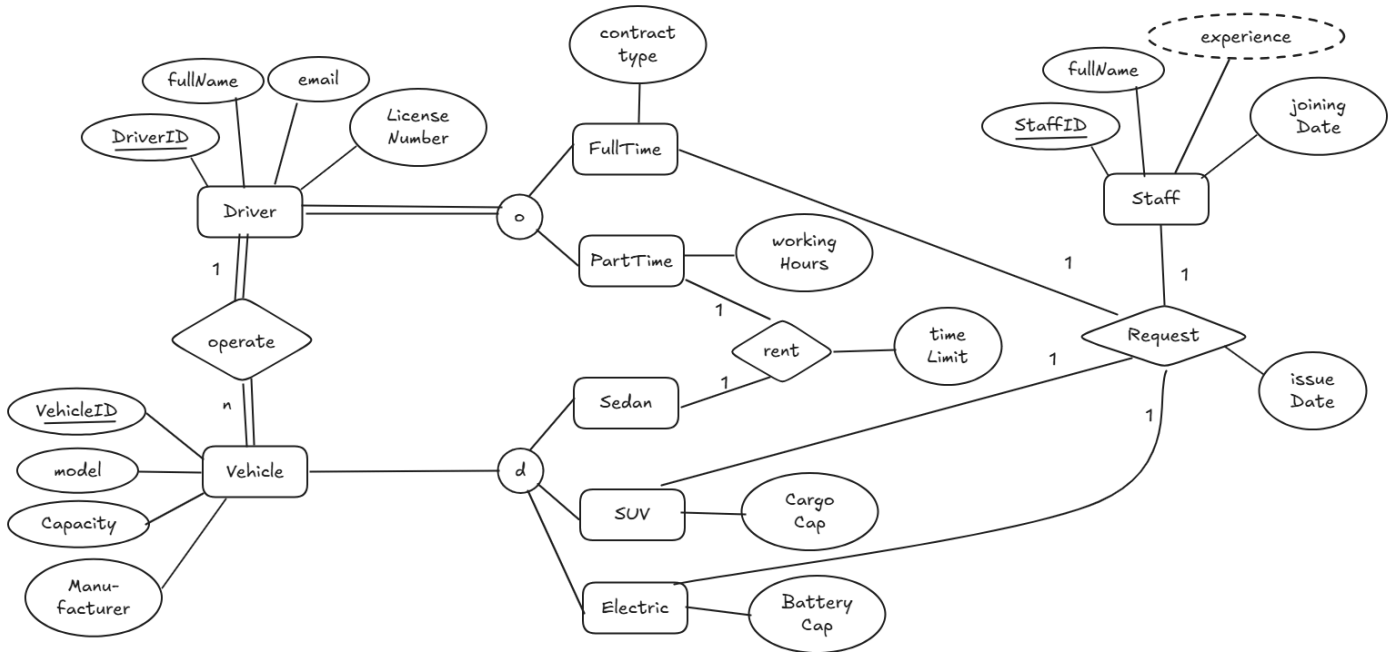
Movies				Actor		
<u>MovieID</u>	MovieName	MovieRating	Genre	<u>ActorID</u>	Actor	workedOn
M02	Mission Impossible	3.5	Action	TC1	Tom Cruise	M02
M12	It	3.1	Horror	FW3	Finn Wolfhard	M12
M14	Rush Hour	4.5	Comedy	JC5	Jackie Chan	M14

- a) Insert <"M12", null, 'X', "Thriller"> into the Movies table.
Identify all constraints which will be violated by this operation and **explain** how you will **solve** it.
- b) Insert <null, "Chris Evans", "M13"> into the Actor table.
Identify all constraints which will be violated by this operation and **explain** how you will **solve** it.
- a) Ans: Key Constraint violation for M12. Solution: set movieID as PK
Domain Constraint violation for 'X'. Solution: set proper datatype
- b) Ans: Entity Integrity Constraint violation for null. Solution: set ActorID as PK
FK Constraint violation for M13, Solution: set proper FK and PK.

Q2. [8 Marks] You are tasked with designing a Driver Management System. The system must meet the following requirements:

- Drivers will have a unique DriverID, full name, email, and license number.
- Each vehicle has a unique VehicleID, model, capacity, and manufacturer.
- Each driver can operate many vehicles, but each vehicle can only be operated by one driver at a time.
- There are only 2 types of drivers: Full-time and Part-time. A driver can be both at the same time. Full-time drivers have a contract_type, while part-time drivers have working_hours.
- Some Vehicles are categorized by their type, such as Sedan, SUV or Electric. A vehicle can belong to only one type. Electric vehicles have BatteryCapacity, SUVs have cargoCapacity.
- A Part-time Driver can rent one Sedan at a time. When a Sedan is rented, the timeLimit is stored.
- A Full-time Driver can request repair from Staff on an SUV or an Electric. The issueDate is recorded when a repair request is made.
- A Staff will have StaffID, FullName, JoiningDate. The experience of a Staff can be calculated from JoiningDate.

Now, draw an **EER** (Extended Entity Relationship) diagram of this system.
Write down all assumptions (if any).



Q3. [8 Marks] Construct a relational Schema by mapping the following EER diagram (Figure Q3). The following conditions must be fulfilled when constructing the schema:

- For mapping the 1:N or 1:1 relationships, you **MUST** use the foreign key approach.
- For mapping the superclasses/subclasses, **select** the applicable option between 8B or 8C and **explain** briefly why the unselected option is not applicable for the given EER.

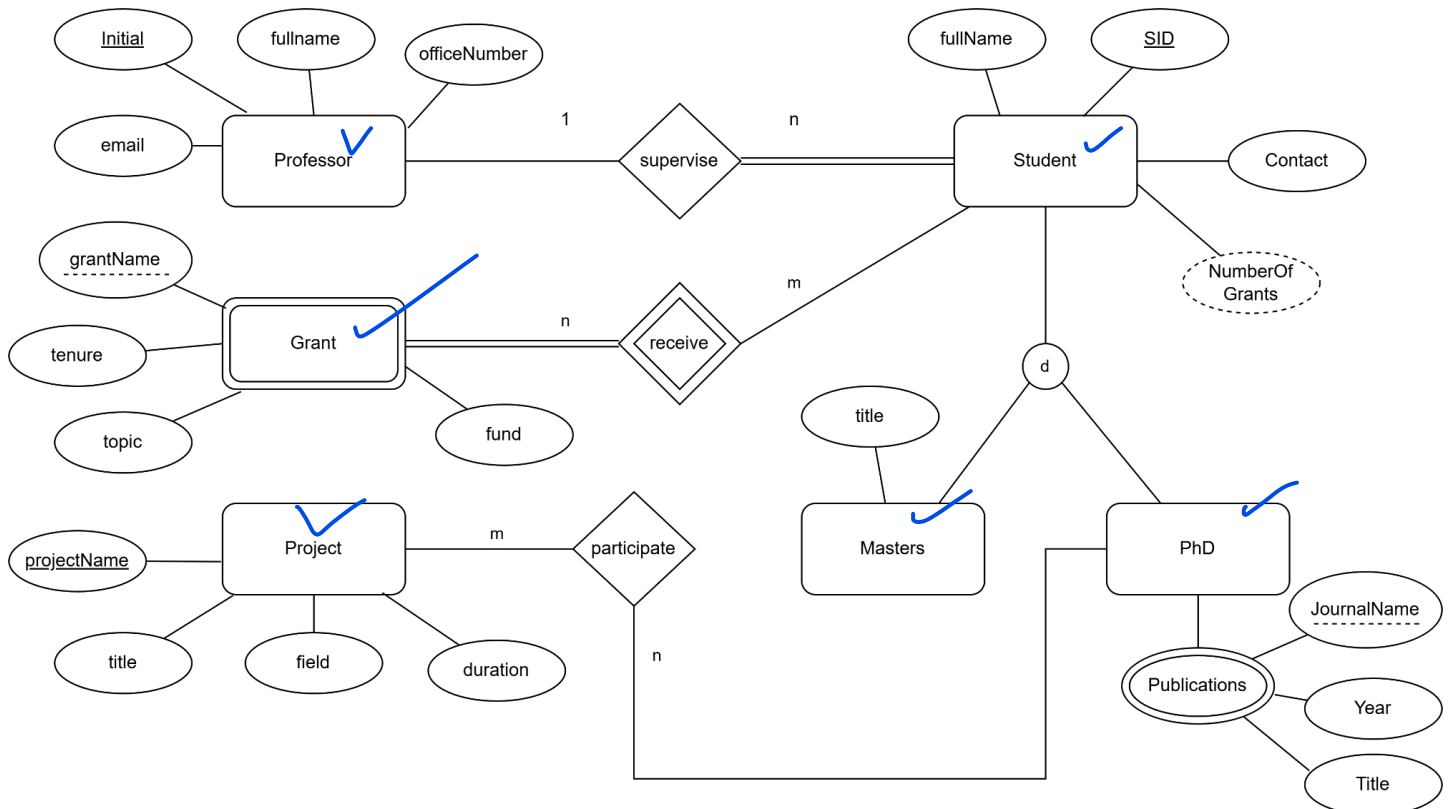


Figure: Q3

Selected 8C

8B can only be used for total.

Professor ✓

<u>Initial</u>	fullName	officeNumber	email
----------------	----------	--------------	-------

Project ✓

<u>ProjectName</u>	title	duration	field
--------------------	-------	----------	-------

Student

<u>SID</u>	fullName	Contact	type	title	initial
------------	----------	---------	------	-------	---------

PhdStudentPublications

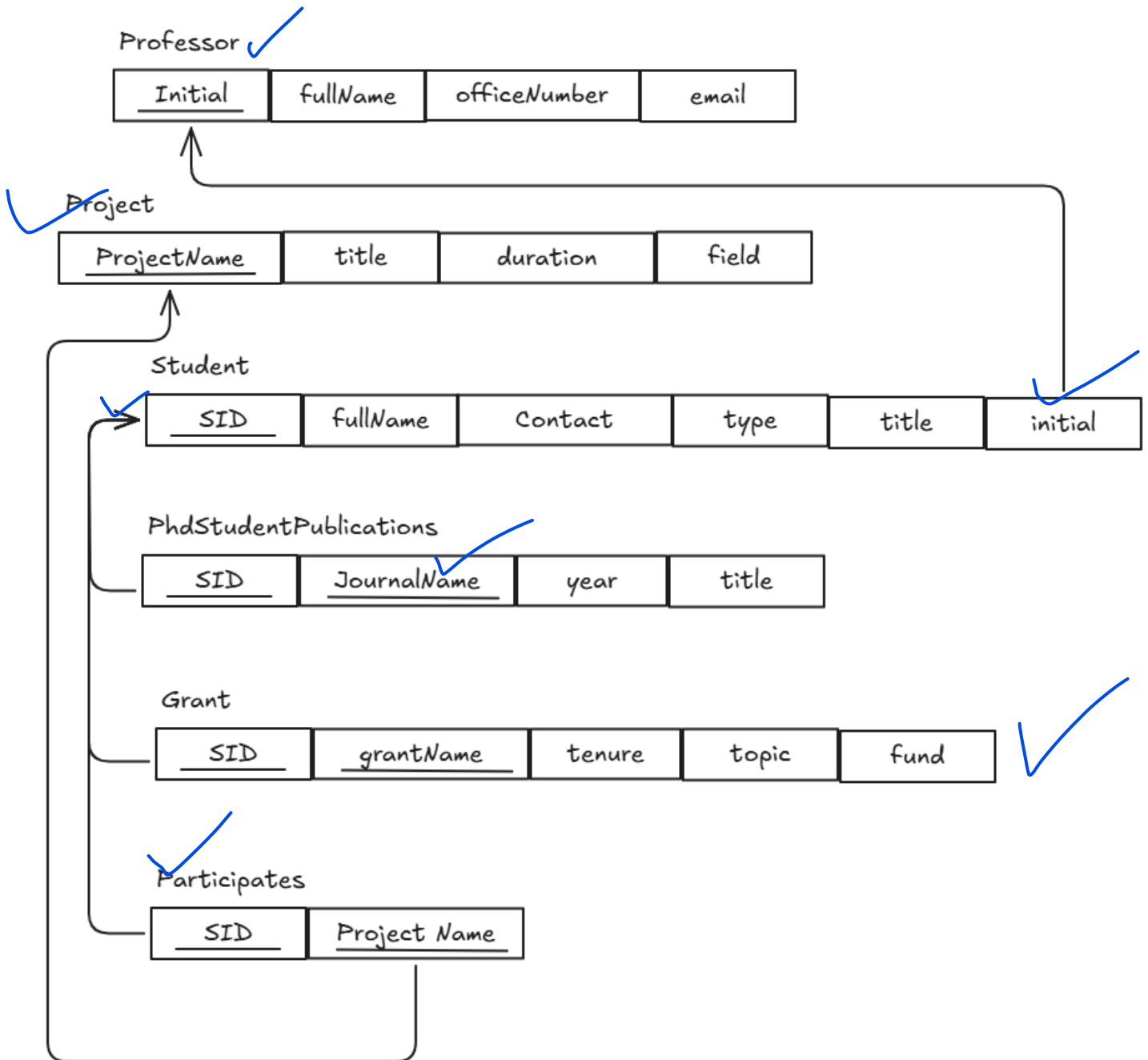
<u>SID</u>	<u>JournalName</u> ✓	year	title
------------	----------------------	------	-------

Grant

<u>SID</u>	<u>grantName</u>	tenure	topic	fund
------------	------------------	--------	-------	------

Participates ✓

<u>SID</u>	<u>Project Name</u>
------------	---------------------



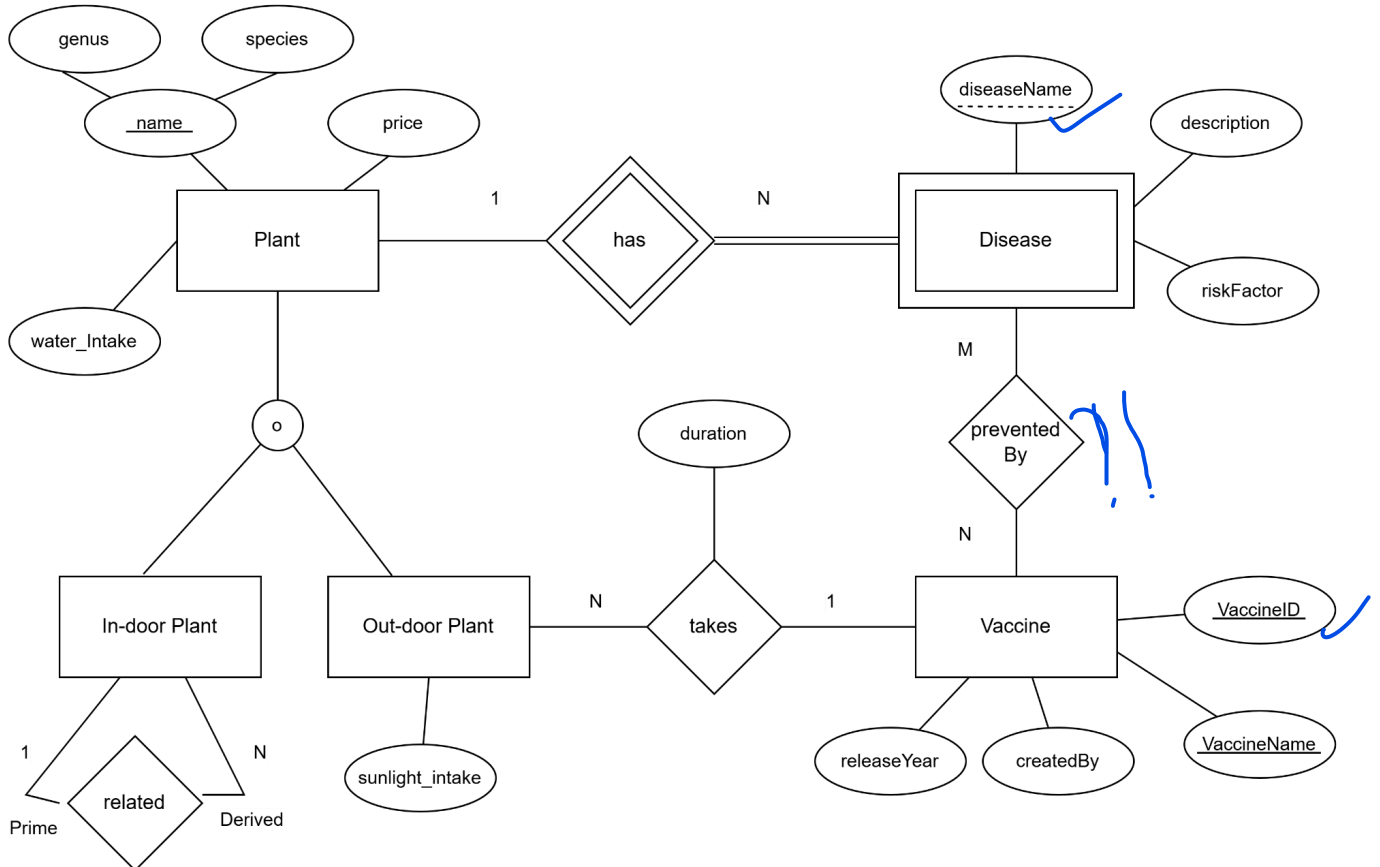
Quiz 2 - Cancelled

Quiz 2 - Cancelled

Question 1 (10 Points)

Construct a relational Schema by mapping the following EER diagram (Figure Q1). The following conditions must be fulfilled when constructing the schema:

- For mapping the 1:N or 1:1 relationships, you **MUST** use the foreign key approach.
- For mapping the superclasses/subclasses, **select** the applicable option between 8C or 8D and **explain** briefly why the unselected option is not applicable for the given EER.



NOTE

8D selected, 8C not applicable for overlapping.

